

Two-Stage PCB Defect Detection System: Binary Screening + 6-Class Classification

Research Project: Semantic Aware Tiny Defect Detection in PCB using CNN

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Abstract

This report evaluates a two-stage convolutional neural network for PCB defect detection. Stage 1 is a binary classifier (defect vs. normal) with a VGG16 backbone, designed to achieve high recall for defects. Stage 2 is a VGG16-based 6-class classifier (open, short, mousebite, spur, copper, pin_hole) that operates only on patches classified as defects. The system was tested on 2,418 patches (1,514 normal, 904 defect). Stage 1 achieved 82% overall accuracy, with normal recall = 0.93, defect recall = 0.63 – indicating that 37% of true defect patches are missed at the first stage. However, Stage 2 performed excellently on the defect patches that passed Stage 1, reaching 96% accuracy with balanced precision/recall across all six defect types. The bottleneck is defect recall in Stage 1; improving this would unlock the full potential of the high-performing Stage 2 classifier.

1 System Overview

The two-stage pipeline processes a 128×128 patch extracted from a PCB image:

1. **Stage 1 (Binary):** Classifies the patch as *defect* or *normal* using a VGG16 model with a sigmoid output. Training emphasised high recall for defects by assigning a class weight of 5 to the defect class.
2. **Stage 2 (6-Class):** Only patches predicted as *defect* are passed to a second VGG16 model that outputs probabilities for the six defect types.

The test set comprised 2,418 patches (1,514 normal, 904 defect) extracted from the DeepPCB dataset.

2 Stage 1 – Binary Defect vs. Normal

Table 1 shows the classification metrics for Stage 1.

Table 1: Binary classification (defect vs. normal) – Test set

Class	Precision	Recall	F1-score	Support
normal	0.81	0.93	0.87	1514
defect	0.85	0.63	0.73	904
accuracy			0.82	2418
macro avg	0.83	0.78	0.80	
weighted avg	0.83	0.82	0.82	

2.1 Observations

- **High normal recall (0.93):** Only 7% of normal patches are falsely flagged as defects – good specificity.
- **Low defect recall (0.63):** Critically, 37% of true defect patches are missed (false negatives). This is the current bottleneck.
- **Precision for defect (0.85):** Among patches predicted as defect, 85% are truly defective. The remaining 15% are false positives (normal patches mistaken as defects).
- **Overall accuracy 82%** is misleading – the system is biased toward predicting normal (because normal class is larger), which inflates accuracy but hides the poor defect recall.

2.2 Why is defect recall low?

Despite using a class weight of 5 for defects during training, the model still struggles to recognise certain defects – especially short circuits, mousebites, and spurs. These may be visually subtle or resemble normal variations. The 128×128 patch size might also be too small to capture the full context of some defects.

3 Stage 2 – 6-Class Defect Classification

Only the 904 defect patches that entered Stage 2 (from the test set) are evaluated here. Table 2 shows the per-class performance.

Table 2: 6-class defect classification on true defect patches (N=1514 patches that passed Stage 1)

Class	Precision	Recall	F1-score	Support
open	0.95	0.98	0.96	305
short	0.97	0.94	0.96	243
mousebite	0.95	0.95	0.95	310
spur	0.94	0.96	0.95	237
copper	1.00	0.98	0.99	211
pin_hole	0.98	0.98	0.98	208
accuracy			0.96	1514
macro avg	0.96	0.96	0.96	
weighted avg	0.96	0.96	0.96	

3.1 Observations

- **Excellent overall performance:** 96% accuracy, all F1-scores ≥ 0.95 .
- **All defect types are well discriminated** – no single class dominates errors.
- **This demonstrates that the visual features for distinguishing defect types are highly discriminative.** The bottleneck is not the 6-class classifier but the binary screening that precedes it.

4 Combined System Performance – End-to-End

In a real deployment, a patch would first be screened by Stage 1. If Stage 1 says *normal*, the system outputs “normal”. If Stage 1 says *defect*, Stage 2 then produces a defect type.

Because Stage 1 misses 37% of true defects, those defect patches never reach Stage 2. Therefore the ****end-to-end recall for defects**** is $0.63 \times 0.96 = 0.60$ (Stage1 recall multiplied by Stage2 accuracy). That is, only 60% of all true defects are correctly classified to the correct type. The remaining 37% are missed entirely (false negatives), and 3% are misclassified as the wrong defect type.

5 Recommendations for Improvement

To make the two-stage system viable for industrial PCB inspection, defect recall in Stage 1 must be increased to at least 0.90. Suggested actions:

1. **Increase patch resolution** to 224×224 (or 256×256) – tiny defects (e.g., short circuits) may become more visible.
2. **Re-balance training data** – further increase the class weight for defects (e.g., from 5 to 10) or use oversampling of defect patches.
3. **Use a more powerful backbone for Stage 1** – e.g., ResNet50 or EfficientNet, which might capture subtle patterns better.

4. **Lower the decision threshold** for Stage 1 (e.g., from 0.5 to 0.3) – this will increase recall at the cost of precision (more false positives). Since Stage 2 is very accurate, false positives can be safely handled.
5. **Consider a single-stage detector** (e.g., YOLO) that directly outputs class probabilities without a separate binary screening step. This may avoid the recall bottleneck altogether.

6 Conclusion

The two-stage system achieves outstanding 6-class classification accuracy (96%) on defect patches that pass the binary screen. However, the binary stage’s low recall (0.63) severely limits the overall effectiveness. Improving defect recall in Stage 1 is the highest priority. Once recall reaches ≥ 0.90 , the complete system would deliver reliable defect detection and classification.